Conditions for the Generation of Beta Oscillations in the Subthalamic Nucleus-Globus Pallidus Network

Paper: Nevado Holgado, Terry and Bogacz (2010). Delayed population-rate modelling study.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

A concrete delayed nonlinear model and conditions for oscillation generation. Relevant to the notebook's model class; its mechanistic predictions require separate behavioural validation.

How does it answer the general questions?

Model mapping. With rates r_S, r_G and constant cortical/striatal drives,

$$\tau_S \dot{r}_S = -r_S + F_S(-w_{GS} r_G(t - T_{GS}) + w_{CS}\mathrm{Ctx}),$$
$$\tau_G \dot{r}_G = -r_G + F_G(w_{SG} r_S(t - T_{SG}) - w_{GG} r_G(t - T_{GG}) - w_{XG}\mathrm{Str}).$$

Take $x = r - r^*$, $a_i = -1/\tau_i$, $b^x = 0$, and signed nonlinear input rows $(0, -w_{GS})$, $(w_{SG}, -w_{GG})$. Set $f_i(v) = [F_i(q_i^* + v) - F_i(q_i^*)]/\tau_i$. The reported sigmoid has $0 \le F_i' \le 1$, giving global slope and sector bounds $[0, 1/\tau_i]$ after centring (derived from their Eq. 3).

Parameters. $\tau_S = 6$, $\tau_G = 14$, $T_{SG} = T_{GS} = 6$, $T_{GG} = 4$ in ms. With $F_i(q) = M_i/[1 + e^{-4q/M_i}(M_i - B_i)/B_i]$, use $(M_S, B_S, M_G, B_G) = (300, 17, 400, 75)$ and $(\mathrm{Ctx}, \mathrm{Str}) = (27, 2)$ in spikes/s. The weight vector $(w_{SG}, w_{GS}, w_{GG}, w_{CS}, w_{XG})$ is $(19, 1.12, 6.60, 2.42, 15.1)$ for the healthy fit and $(20, 10.7, 12.3, 9.2, 139.4)$ for the diseased fit (Tables 1, 3; PDF pp. 2–4). Their pulse experiments use a fitted additive electrode-current term in the STN equation (Eq. 5); no LFP observation map is identified.

Interpretation. Delays, coupling and external drive determine oscillation onset within this model. Predictions about removing beta are model-level results, not proof that beta suppression alone treats symptoms (PDF pp. 5–12).

What papers does it cite that could provide more information?

- Plenz and Kitai (1999), *A basal ganglia pacemaker formed by the subthalamic nucleus and external globus pallidus*: experimental loop hypothesis (PDF p. 13).
- Terman et al. (2002), *Activity patterns in a model for the subthalamopallidal network of the basal ganglia*: alternative spiking mechanism (PDF p. 13).